Other lifestyle modifications

Physical activity – Regular aerobic exercise like brisk walking can help to strengthen the heart muscle, promote weight loss or help maintain weight, raise HDL cholesterol, improve blood glucose levels and reduce blood pressure.

Smoking and stress – Refraining from smoking and appropriate strategies for coping with stress would help to prevent heart disease.

Case study

Kumar is a 60 year old executive who is obese. He smokes and occasionally consumes alcohol. He is very ambitious and does not like to do exercise. He has a strong family history for heart disease.

1. What are his risk factors?
2. What lifestyle modification will you advice him?

1.6.2 Hypertension

Hypertension is the number one killer disease in India, with a higher prevalence among the urban adult population.

The normal pressure of blood against the walls of the arteries is called blood pressure or hypertension. The normal blood pressure, when the heart contracts, is called systolic pressure (SBP) and is between 100-140 mm of mercury. When the heart relaxes, the pressure is known as diastolic pressure (DBP) with values of 70-90 mm of mercury.

Hypertension in adults aged 18 years and older is defined as systolic blood pressure of 140 mm Hg or greater and/or diastolic blood pressure of 90 mm Hg or greater.

It is also called "the silent killer" because it does not exhibit any signs or symptoms to indicate its presence. When undetected, untreated or uncontrolled can lead to serious consequences like stroke.

Hypertension may be classified as

- **Primary or essential hypertension** - It refers to the condition where the specific cause is unknown. It involves a complex interaction between poor lifestyle choices and genes. The lifestyle factors include a poor diet – high sodium, low fruit and vegetable intake, obesity, smoking, lack of exercise and stress.
- **Secondary hypertension** - This has a known cause and may arise as a result of another disease. E.g. Individuals with kidney disease have secondary hypertension.

Classification of blood pressure

Category	Systolic (mmHg)	Diastolic (mmHg)
Optimal	<120	And <80
Normal	<130	<85
High normal	130-139	85-89
Hypertension		
Stage 1	140-159	90-99
Stage 2	160-179	100-109
Stage 3	≥180	≥110

Source: Indian Guidelines for management of Hypertension II, 2013

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